

Answer-Based Entity Extraction and Alignment for Visual Text Question Answering

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ABSTRACT

As a variant of visual question answering (VQA), visual text question answering (VTQA) provides a text-image pair for each question. Text utilizes named entities to describe corresponding image. Consequently, the ability to perform multi-hop reasoning using named entities between text and image becomes critically important. However, existing models pay relatively less attention to this aspect. Therefore, we propose Answer-Based Entity Extraction and Alignment Model (AEEA) to enable a comprehensive understanding and support multi-hop reasoning. The core of AEEA lies in two main components: AKECMR and answer aware predictor. The former emphasizes the alignment of modalities and effectively distinguishes between intra-modal and inter-modal information, and the latter prioritizes the full utilization of intrinsic semantic information contained in answers during training. Our model outperforms the baseline by 2.24% on test-dev set and 1.06% on test set, securing the third place in VTQA2023(English).

CCS CONCEPTS

• **Computing methodologies** → **Computer vision**; **Natural language processing**.

KEYWORDS

Visual Text Question Answering Challenge (VTQA 2023), answer semantics, entity extraction, feature alignment

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1 INTRODUCTION

Transformers [21] unifies multimodal data representation, enabling researchers to excel in complex tasks like image-text retrieval, visual entailment, visual question answering (VQA) and visual reasoning. Among these tasks, VQA demonstrates strong practical value. As a fine-grained large-scale classification task, it also presents significant challenges.

VQA task encompasses various variants, and in this work, we specifically focus on VTQA [4]. As shown in Figure 1, VTQA stands for a form of VQA, providing an image-text pair for each question. Text describes corresponding image using named entities, and then these entity names are used to formulate questions. The questions are categorized into three types: Yes/No (YN), Extraction (E), and Generation (G).

There are many challenges for VTQA. Firstly, in terms of image and text embeddings, current approaches [6, 7, 12] often lack specially designed feature alignment techniques. Instead, they primarily rely on a linear projection to map these modalities into a shared semantic space. Secondly, it is crucial to handle the entities in text and align them with the corresponding image and question. Finally, it is worth noting that the answer set within the VTQA

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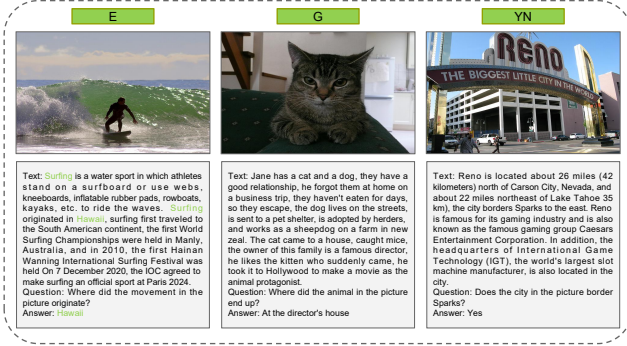


Figure 1: The VTQA dataset. Based on the types of answers, questions can be categorized into three types: Yes/No (YN), Extraction (E) and Generation (G).

dataset encompasses a significant reservoir of semantic information, which is frequently underutilized in existing methods.

Considering the aforementioned limitations, we propose AEEA. We use Aligned Key Entity Extraction (AKEE) to align image and text embeddings and extract entities. Then we utilize Unified Attention Cross-Media Reasoning (UCMR) to align entities from various modalities. AKEE and UCMR collaborate to form Aligned Key Entity Cross-Media Reason (AKECMR). In order to capture answer semantics during training process, we introduce answer aware predictor. By incorporating these components, AEEA effectively addresses the aforementioned challenges.

In summary, our contributions are as follows:

1. We propose AKEE to tackle the challenges of feature alignment and entity extraction. We first align image and text features, and then extract entities from them to answer questions. By adopting this approach, we can obtain high-quality entities and perform multi-hop reasoning smoothly.
2. We introduce UCMR for entity alignment. It explicitly considers the distinctions between intra-modal and inter-modal information, dynamically controlling the representations of modalities based on different interactions.
3. We introduce the answer aware predictor, which incorporates answer semantics during the training process. It aids the predictor in acquiring the semantic information pertaining to the answer, thereby optimizing the overall quality of the predictor.

These contributions aim to address the challenges related to feature alignment, entity extraction and effective utilization of answer semantics in VTQA.

2 RELATED WORK

Vision-and-Language Pre-training (VLP) models. Currently, a popular paradigm for addressing VQA task is to treat them as one of downstream tasks of VLP models. The process of pre-training involves endowing the model with a substantial volume of data and employing a range of loss functions, such as image-text matching loss, image-text contrastive loss, and masked language modeling loss. After pre-training, a substantial portion of the model weights is fixed, while classification head is introduced for fine-tuning on the VQA dataset.

There are two main architectures in VLP models: single-stream and dual-stream. In single-stream models [1, 3, 5, 8, 13, 16], the image and text features are concatenated and fed into the network, resulting in a single feature vector as the output. In dual-stream models [15, 17, 20], the text and images are processed separately by feeding them into two different networks. The features are then interacted through cross-attention, and the model generate individualized feature vectors for text and images.

Independent VQA models. Compared to VLP models, independent VQA models [18, 19, 24] exhibit a stronger focus on specific targets. The images and questions are initially embedded separately to obtain their respective representations, and then fed into feature-interaction layers for training. Finally, the features are concatenated and passed through a classification head for results prediction. Based on different methodologies employed for extracting image features, independent VQA models can be categorized into two distinct types: region-feature-based models and grid-feature-based models. There is also scenario where both types of features are interchangeable [4].

Region features are extracted through the utilization of object detection networks, which are frequently pre-trained on the Visual Genome dataset [14]. These networks employ non-maximum suppression to effectively identify and retain a maximum of 100 bounding boxes. Subsequently, the region features are derived from these bounding boxes. Many works [2, 11, 25] demonstrate the effectiveness of region features. Grid features directly use backbone network [9, 22] to extract image features. For instance, in the work of Jiang et al. [10], feature maps are generated as grid features for images. Notably, within these feature maps, distinct locations are associated with different regions in the image. We conduct an in-depth investigation of such models, achieving favorable results even without the utilization of multimodal large language models

3 METHODOLOGY

In this section, we commence by providing an overarching presentation of the AEEA structure, followed by entailed description of different modules. AEEA is built on the foundation laid by [4], improves Key Entity Extract Layer (KEE) and Cross-Media Reason Layer (CMR), and introduces answer aware predictor that utilizes answer semantics.

3.1 Overview

As depicted in Figure 2, AEEA primarily comprises three essential components: feature representation, AKECMR and answer-aware predictor.

Firstly, we generate appropriate feature representations for the input text and images. We utilize GloVe, which is pre-trained on a large-scale corpus, to convert text and questions into word vectors. These word vectors are then fed into a one-layer LSTM network to learn their representations. Regarding images, features are extracted using Faster R-CNN trained on the Visual Genome dataset [14]. Each region of interest containing detected objects is encoded as a [1, 2048] embedding.

Next, AKECMR is employed to extract and align multimodal entities. AKECMR consists of two key components: AKEE and UCMR. AKEE aligns image and text features and extract their entities. 32

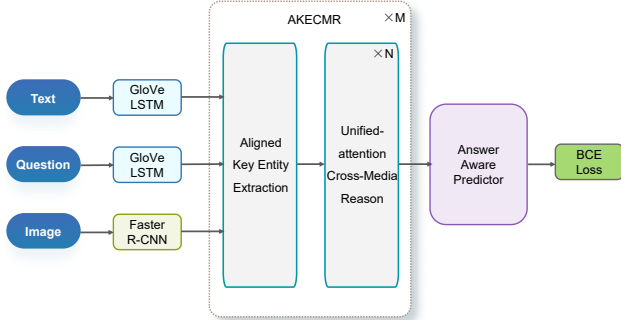


Figure 2: The architecture of AEEA. We extract and refine entities using AKECMR, and leverage answer semantics within the answer-aware predictor to make predictions for the final results.

entities are extracted for each image and text, while 24 entities are extracted for each question. Each entity has a dimension of [1, 2048]. Subsequently, these entities are concatenated and input to UCMR. UCMR explicitly distinguishes between intra-modal and inter-modal information, further aligning the extracted multimodal entities.

Finally, the key entities derived from AKECMR serve as input for the predictor. When predicting results, we introduce answer aware predictor to harness the power of answer semantics.

3.2 Aligned Key Entity Extraction

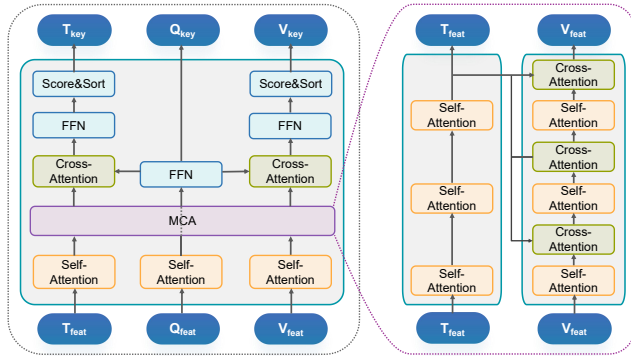


Figure 3: The structure of AKEE. The primary objective of AKEE is to facilitate the alignment of multimodal information and conduct entity extraction.

To address the issue of modality alignment and entity extraction, we propose AKEE, which incorporates the Modular Co-Attention (MCA) [25] into KEE. In KEE, the feature representations of images, text, and questions undergo linear projection and self-attention, which are then directly used for entity extraction. However, the presence of semantic disparities between image and text impedes the attainment of precise entity extraction. Therefore, we incorporate MCA before extracting key entities, to facilitate interaction between image and text features and enhance semantic alignment.

As shown in Figure 3, MCA consists of self-attention modules and cross-attention modules, employing an encoder-decoder architecture. The text is fed into the encoder, while the image is passed into the decoder. Following a sequence of three self-attention layers, the textual information produces outputs that serve as keys and values during the computation of cross-attention with the image features. Effective alignment between the image features and text features is accomplished by integrating MCA, thereby augmenting the performance of entity extraction.

3.3 Unified Attention Cross-Media Reasoning

We enhance CMR and introduce UCMR, as depicted in Figure 4. In CMR, entities originating from multiple modalities are concatenated and subjected to a self-attention module for processing. Nevertheless, this methodology inadvertently disregards intra-modal and inter-modal interaction. To rectify this limitation, we propose the incorporation of parameter matrices within the self-attention to explicitly modulate the output.

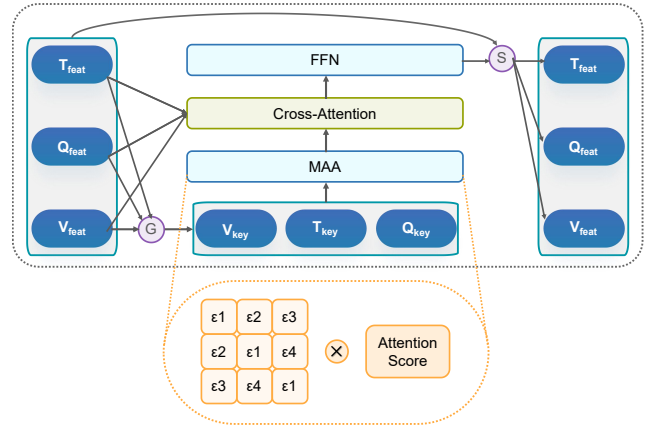


Figure 4: The structure of UCMR. It is built upon the CMR model with the addition of MAA.

Taking inspiration from [23], we adopt a perspective wherein image, text, and question are regarded as distinct forms of information. Subsequently, we concatenate them together as input to the self-attention module, where Modal Adaptive Attention (MAA) is utilized to adjust the output. This approach explicitly differentiates between intra-modal and inter-modal information, enabling dynamic control over the representations of modalities under varying interaction contexts.

Specifically, assuming that V , T , and Q represent image, text and question respectively. We apply AKEE to obtain the corresponding entities V_{key} , T_{key} and Q_{key} . Subsequently, these representations are concatenated into a unified entity matrix $G = [V_{key}, T_{key}, Q_{key}] \in \mathbb{R}^{(n_v+n_t+n_q) \times d}$. This resulting matrix is then fed into UCMR. The formulation can be described as follows:

$$U_{att} = MAA(G, n_v, n_t, n_q) \quad (1)$$

$$= softmax\left(\frac{GG^T}{\sqrt{d_G}}\right) \cdot W \cdot G \quad (2)$$

$$W = \begin{bmatrix} [\epsilon_1]_{n_v \times n_v} & [\epsilon_2]_{n_v \times n_t} & [\epsilon_3]_{n_v \times n_q} \\ [\epsilon_2]_{n_t \times n_v} & [\epsilon_1]_{n_t \times n_t} & [\epsilon_4]_{n_t \times n_q} \\ [\epsilon_3]_{n_q \times n_v} & [\epsilon_4]_{n_q \times n_t} & [\epsilon_1]_{n_q \times n_q} \end{bmatrix} \quad (3)$$

n_v , n_t and n_q represent the number of key entities in V_{key} , T_{key} and Q_{key} . W is composed of trainable parameters ϵ_1 , ϵ_2 , ϵ_3 , ϵ_4 .

After applying MAA to obtain new unified representations, cross-attention is modeled between U_{att} and the original features $[V_{feat}, T_{feat}, Q_{feat}]$. In AKEE, the entities are directly indexed from the original features and later reinstated to their initial positions in UCMR. The cross-attention models the alignment of the updated entities with original features. The resulting updated V_{feat} , T_{feat} , and Q_{feat} are subsequently employed as input for the subsequent layer of AKECMR.

3.4 Answer Aware Predictor

Through AKECMR, we obtain key entities for answer prediction. In the predictor, we adopt the structure from [25], as shown in Figure 5. Similar to traditional VQA algorithms, they treat VQA as a multiclass classification problem and represent each answer using one-hot encoding. However, this approach clearly sacrifices a significant amount of semantic information inherent in the answers, leading to suboptimal results. In light of this, we integrate the semantic features of answers into the predictor, thereby engendering an answer aware predictor.

Assuming the presence of a training sample, consisting of a question-answer pair with the following content: "What color is Jack's hat?" and "White". The answer "White" corresponds to the index 43. The conventional approach disregards the inherent semantic information encapsulated within the word "White" itself and solely relies on the numerical value of 43. How can we rectify this issue? We opt to rephrase the question based on the answer, thereby transforming it into a declarative sentence such as "Jack's hat is white." This can be readily accomplished through the utilization of a language model. We perform this transformation during training with a probability of 50%.

By substituting the original question with the rephrased declarative sentence, the model is effectively informed about the conclusive answer. Consequently, AKECMR effectively captures highly accurate entities, which are denoted as cheating entities. Directly acquiring the answer during inference is unattainable. Nonetheless, these cheating entities play a crucial role as guiding elements, enables the predictor to more accurately predict the answer. Despite the unavailability of cheating entities during inference, the semantic information conveyed to the predictor during training still enhances prediction accuracy.

4 EXPERIMENTS

4.1 Dataset

We conducted our experiments on the VTQA dataset [4]. In contrast to VQA, VTQA incorporates textual information to describe images using named entities and subsequently utilizes these named entities to formulate questions. The dataset is partitioned into four distinct subsets: train, validation, test-dev, and test. The specific sizes of each split are as follows: 11,312 samples for training, 1,245 samples

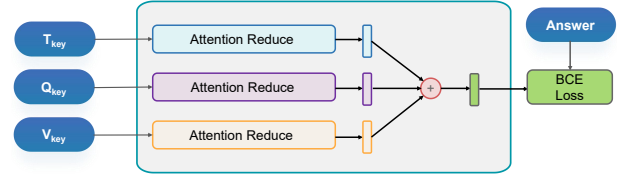


Figure 5: The structure of answer aware predictor. There is a 50% probability of utilizing the answer to rephrase Q_{key} as a declarative statement.

for validation, 2,189 samples for test-dev, and 9,035 samples for final test.

4.2 Main Results and Ablation Study

We conduct ablation experiments on the test-dev set to demonstrate the effectiveness of three modules proposed in our approach. Training is conducted on 2 GPUs of V100 and Table 1 presents the results. Comparing to the baseline [4], UCMR demonstrates a 0.51% improvement in exact match (EM), while the incorporation of AKEE yields a 1.74% increase. Moreover, the introduction of answer aware predictor contributes to an overall EM performance enhancement of 1.97%. By means of model ensemble, the final EM attains 0.6062, reflecting a notable increase of 2.24%. On the test set, our method yields a 1.06% improvement. The results demonstrate the effectiveness of our method in tackling VTQA task.

	EM (test-dev)	EM (test)
Baseline	0.5838	0.5760
UCMR	0.5889	0.5763
UCMR + AKEE	0.6012	0.5826
UCMR + AKEE + Answer Aware Predictor	0.6035	0.5844
Model Ensemble	0.6062	0.5866

Table 1: Experimental results. All the results are computed on the VTQA2023(English).

5 CONCLUSION

We propose AEEA, a VTQA training paradigm that integrates answer semantics. AEEA explicitly harnesses answer semantics during training to enhance feature representation. Additionally, we introduce AKECMR, which incorporates modality alignment to improve the quality of extracted entities. These modules are plug-and-play, facilitating effortless integration into a wide range of multimodal tasks. Notably, we believe that these methods hold promising implications for other VQA tasks.

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